

Team 1 - COLX 585*

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1 Task Description

The task in this project is to perform a membership inference attack (MIA) on finetuned language models. More specifically, we aim to predict whether a given sample was included in the models' finetuning dataset by assigning each sample a membership score. This task is important because it helps evaluate whether a model leaks information about its training data, which is closely related to privacy and memorization risks in machine learning systems. Therefore, the goal of this project is to develop a method that can distinguish member samples from non-member samples as accurately as possible and submit these predictions to the Kaggle competition.

2 Background

2.1 High-Level Description

The 'UBC-SLIME/colx585_group_project_data' dataset is designed and a continuation of our Membership Inference Attacks (MIA), but with the addition of Vision-Language Models. Each of the samples we have viewed on 'milestone4/document/data_inspection.md' consists of five features: a unique 'id', an 'image' (stored as raw bytes), the target 'is_member' binary label (0 or 1), a 'text' string containing a conversational prompt and response, and a 'type' category (e.g., 'seen_img_unseen_txt'). The 'type' feature is particularly interesting and useful as it seems to categorize if the image, text, or both has been seen during training which may be very useful.

2.2 Literature Review - Milestone 4

2.2.1 Paper 1: Membership Inference Attacks against Large Vision-Language Models

This paper studies privacy risks in large vision-language models (VLLMs), especially the possibility that sensitive training data such as private photos or medical records may be exposed through membership inference attacks. The authors note that this problem remains challenging because there is a lack of suitable datasets and evaluation methods. To address this, they introduce a benchmark for

training data detection in VLLMs, propose a token-level image MIA pipeline, and design a new attack metric.

Main Task. The main task is to detect whether a specific image or text sample was used to train a vision-language model. This is formulated as a binary classification problem, where the attacker determines whether a target sample is a member or non-member of the training set.

Dataset. The authors construct **VL-MIA**, which contains three sub-datasets: **VL-MIA/DALL-E** (592 member images from LAION-CCS and DALL-E-generated non-member images), **VL-MIA/Flickr** (600 MS COCO member images and recent Flickr photos as non-members), and **VL-MIA/Text** (600 instruction-tuning text samples and GPT-4-generated non-member answers).

Models. The paper evaluates three open-source VLLMs: **LLaVA 1.5**, **MiniGPT-4**, and **LLaMA-Adapter V2**. It also reports results on closed-source **GPT-4** (vision-preview API).

Main Contribution. The main contribution is the first MIA benchmark designed specifically for VLLMs. The paper also proposes a **cross-modal pipeline** that performs image MIA using text logit slices, which is important because image tokens are not directly accessible in VLLMs. In addition, the authors introduce **MaxRényi-K%**, a target-free metric based on Rényi entropy that can be applied to both image and text modalities.

2.2.2 Paper 2: The Sample Complexity of Membership Inference and Privacy Auditing

This paper studies membership inference attacks (MIAs) from a theoretical perspective by asking how much auxiliary information an attacker needs in order to succeed. Instead of focusing directly on LLMs or VLMs, it studies a simpler setting called Gaussian mean estimation, where a model learns from n samples and releases a noisy estimate. The main result shows that an attacker may need far more reference samples than the model used for training, which differs from many existing MIA

*Repository: <https://github.ubc.ca/snalyf/TFC>

methods that only use $O(n)$ samples. The paper also argues that current privacy audits may underestimate risk if they do not fully use information about the data distribution.

Main Task. The paper studies the sample complexity of membership inference attacks. More specifically, it asks how many auxiliary samples from the same population are needed for an attacker to distinguish whether a target example was part of the training set or not. The analysis focuses on sample-based MIAs in Gaussian mean estimation and compares the unknown-covariance and known-covariance settings.

Dataset. This paper does not use a real benchmark dataset such as Flickr, LAION, or WikiMIA. Instead, it considers a theoretical setting where the data come from a d -dimensional Gaussian distribution $N(\mu, \Sigma)$, and the training set consists of n i.i.d. samples drawn from that distribution.

Models. The model studied in this paper is not a VLM or LLM, but a Gaussian mean estimator. The main estimator is the noisy empirical mean, which outputs

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i + \rho Z, \quad Z \sim N(0, \Sigma).$$

The attacker then tries to infer whether a target sample was used in training based on this released estimate and a set of auxiliary samples.

Main Contribution. The paper shows that when the covariance is unknown, a successful sample-based MIA may require much more auxiliary data than expected, up to $\Omega(n + n^2 \rho^2)$ samples. This means the attacker may need far more data than the model used for training. The paper also shows that the problem becomes much easier when the covariance is known, highlighting that estimating covariance structure is the main challenge. Overall, the paper suggests that many current MIA methods, which typically use only $O(n)$ reference samples, may underestimate real privacy risk.

3 Data

3.1 High-Level Description

The UBC-SLIME/colx585_group_project_data dataset serves as a continuation of our work on Membership Inference Attacks (MIA), with a specific focus on Vision-Language Models (VLM). As documented in

milestone4/document/data_inspection.md, each sample in the dataset consists of five primary features:

- **id:** A unique identifier for each sample.
- **image:** The visual component of the sample, stored as raw bytes.
- **is_member:** A binary target label (0 or 1) indicating training set membership.
- **text:** A string containing both a conversational prompt and its corresponding response.
- **type:** A categorical feature (e.g., seen_img_unseen_txt) indicating the specific membership condition of the sample’s components.

The type feature is particularly significant as it allows for granular analysis of how memorization differs when only the image, only the text, or both modalities have been seen during training.

3.2 Descriptive Statistics

Table 1 provides the descriptive statistics for the official dataset. Note that text length calculations are based on the character count of the combined prompt and response string.

Split	N	Mean	Med.	Max	Min	SD
train	6,000	261.93	246	813	98	89.24
validation	1,200	258.05	241	737	104	86.61
test	6,000	247.19	232	1,079	82	91.35

Table 1: Descriptive statistics of text lengths across dataset splits.

4 Contributions

4.1 Milestone 4

Member	Contributions	Percentage
Tianhao Cao	Project Work	25%
Yusen Huang	Project Work	25%
Marco Wang	Project Work	25%
Darwin Zhang	Project Work	25%

Table 2: Team contributions for Milestone 4.